Lexical Leakage and Shortcut Sensitivity in Public Mental-Health Text Classification: An Interpretable Validation Study

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***Abstract*—Background: Public mental-health text datasets are widely used for benchmarking, but high performance can reflect lexical leakage, crisis phrases, length effects, near duplicates, or platform artifacts. Objective: We audit shortcut sensitivity in public mental-health text classification while avoiding diagnostic claims. Methods: We analyzed 49,451 public text samples labeled Normal, Anxiety, Depression, or Suicidal. We evaluated a four-class task and three one-vs-normal tasks using TF-IDF logistic regression, calibrated linear SVM, multinomial naive Bayes, and a secondary random forest. We performed dataset audit, keyword and crisis-term ablations, artifact removal, random deletion controls, length-matched evaluation, near-duplicate-safe splitting, calibration analysis, and lexical feature categorization. Results: Four-class classification reached macro F1 0.776 with TF-IDF + Logistic Regression. The best one-vs-normal AUROCs ranged from 0.984–0.988. Despite strong performance, shortcut sensitivity remained: the strongest overall task/model pair showed length sensitivity 0.425 and was assigned high shortcut risk by a heuristic leakage-audit rule. Conclusion: Classical TF-IDF models predict public mental-health dataset labels strongly, but the evidence is best interpreted as benchmark robustness auditing rather than clinical deployment evidence.**

***Index Terms*—mental-health NLP, shortcut learning, lexical leakage, calibration, interpretable machine learning, health informatics**

I. Introduction

Mental-health text classification is a prominent biomedical informatics research area because public and patient-generated language can contain signals related to distress, anxiety, depression, and suicidal ideation. Public benchmarks are attractive because they are easier to access than protected clinical data, support reproducible model comparison, and allow rapid evaluation of feature representations and classifiers. They are also high-risk evidence sources. Labels in public datasets are not necessarily clinician-confirmed, and the distribution of text may reflect platform, sampling, and annotation artifacts as much as mental-health language.

High benchmark scores are especially difficult to interpret in mental-health AI. A model may learn clinically meaningful symptom language, such as expressions of sleep disturbance or hopelessness, but it may also rely on direct label words, crisis phrases, post length, duplicated examples, URLs, or source-specific tokens. Lexical leakage differs from meaningful evidence because it can encode the dataset construction process rather than a transportable language pattern. This paper studies whether public mental-health text classifiers learn broad symptom-language patterns or exploit direct label words, crisis phrases, text length, near-duplicate examples, and platform artifacts. Figure 1 summarizes the study design. Our focus is not clinical diagnosis. Because the dataset does not provide clinician-confirmed diagnostic labels or detailed sampling provenance, our claims are restricted to dataset-label classification and robustness auditing rather than mental-health diagnosis.

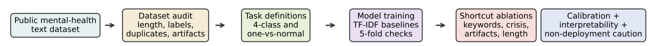

Fig. 1. Study design. A public mental-health text dataset is audited, converted into four task definitions, evaluated with TF-IDF baselines, stress-tested with shortcut ablations, assessed for calibration, and reviewed through lexical feature categories. The final stage is caution against clinical deployment from benchmark evidence alone.

The contributions are:

1) We audit a public mental-health text classification dataset for label, length, duplicate, and lexical artifacts.
2) We compare multiclass and cleaner one-vs-normal tasks rather than relying on a clinically coarse binary mapping.
3) We quantify shortcut sensitivity using direct keyword removal, crisis-term ablation, artifact removal, random deletion controls, length-matched evaluation, and near-duplicate-safe splitting.
4) We evaluate calibration and uncertainty for screening-support research.
5) We provide lexical feature analyses identifying the extent to which top predictors reflect symptom language versus leakage or artifacts.

The intended audience is not only model developers, but also health-informatics researchers who use public datasets as evidence. A benchmark can be useful even when it is not clinically definitive, but only if its assumptions are visible. The central premise of this study is that a defensible report for sensitive health-language classification should describe the label contrast, audit the dataset, test shortcut perturbations, quantify uncertainty, and inspect the lexical evidence used by interpretable models.

II. Related Work

Mental-health NLP has been studied in clinical interviews, social media, online forums, and other text sources. Public text classification benchmarks are useful for reproducible comparison, but they often combine heterogeneous sources and annotation procedures. Consequently, strong classification results can reflect dataset-label regularities rather than stable mental-health constructs.

Clinical-interview datasets provide a different evidence setting. DAIC-WOZ, for example, supports clinical-interview depression-screening research [1]. Even in clinical datasets, PHQ-style instruments are screening labels rather than definitive diagnoses [2]. This distinction is important for interpreting public datasets, where labels may be inferred from posts, communities, or annotations rather than clinical assessment.

Dataset documentation and bias analysis are central to trustworthy health-informatics research. Datasheets for datasets emphasize provenance, collection procedure, intended use, and limitations [3]. Health equity work further shows that model errors can vary across populations and contexts [4]. Although this dataset does not provide demographics for subgroup analysis, the same principle motivates auditing label balance, length, duplicates, and lexical artifacts.

Shortcut learning can undermine medical AI generalization when models exploit site-specific, acquisition-specific, or label-specific artifacts rather than transportable signals [5]. Calibration and interpretability are related safeguards: calibrated probabilities are necessary for risk communication, and feature audits help identify whether a model relies on plausible language or leakage [6], [7]. We apply this validation lens to a public Hugging Face mental-health text dataset [8].

Interpretable linear text models remain valuable in this setting. Although transformer models can improve predictive performance, TF-IDF baselines expose weighted lexical features directly and are computationally reproducible on ordinary hardware. This transparency makes them appropriate for leakage auditing. If a simple linear model achieves high performance using obvious label words or platform tokens, then more complex models trained on the same data may also exploit those cues unless explicitly tested.

III. Dataset and Audit

DAIC-WOZ and E-DAIC files were not available locally, so the experiments used the public Hugging Face *Mental-Health Text Classification Dataset*. The processed dataset contains 49,451 samples: 18,151 Normal, 5,579 Anxiety, 14,517 Depression, and 11,204 Suicidal. The dataset source provides public benchmark labels but does not establish clinician-confirmed psychiatric diagnoses. Therefore, this study evaluates dataset-label prediction and shortcut sensitivity rather than true mental-health status. This distinction is central because public mental-health datasets are frequently used to benchmark models despite uncertain sampling provenance.

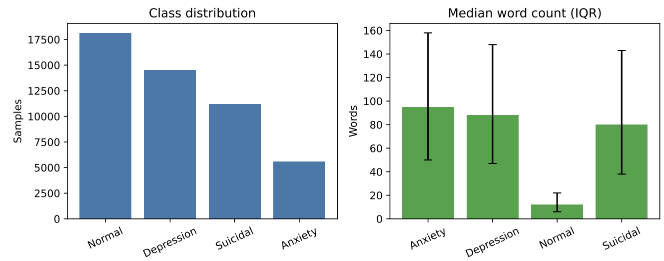

Fig. 2. Dataset audit. The left panel shows class imbalance; the right panel shows median word count with interquartile range by class. Length differences motivate length sensitivity and length-matched analyses.

TABLE I
Dataset audit summary. Prevalence columns are percentages within class.

Class	N	%	Median word count	IQR	Direct keyword prevalence (%)	Crisis keyword prevalence (%)	URL/artifact prevalence (%)
Normal	18,151	36.7%	12.0	16.0	1.4%	0.8%	5.8%
Depression	14,517	29.4%	88.0	101.0	47.9%	14.9%	2.0%
Suicidal	11,204	22.7%	80.0	105.0	32.5%	36.9%	4.1%
Anxiety	5,579	11.3%	95.0	108.0	55.9%	6.0%	2.3%

The audit found zero missing or empty texts, 89 exact normalized duplicates, and 313 approximate near-duplicate pairs using character 5-gram TF-IDF nearest-neighbor similarity with cosine similarity at least 0.92. Text length varied substantially by class: Normal examples were much shorter than the Anxiety, Depression, and Suicidal classes. Direct keyword prevalence also differed by class, with Anxiety and Depression containing direct label words far more often than Normal. Figure 2 and Table I show class counts, word-count differences, and keyword/artifact prevalence.

Exact duplicates were counted after lowercasing and removing non-alphanumeric variation. Approximate near duplicates were estimated with character five-gram TF-IDF because this representation is sensitive to repeated phrasing, boilerplate, and lightly edited posts while remaining scalable for the dataset size. The threshold of 0.92 was treated as a conservative indicator of very high textual similarity rather than broad semantic paraphrase, so the reported near-duplicate count should be interpreted as an audit signal rather than a complete duplicate census.

IV. Task Definitions

Table II defines the final tasks. Task A preserves all four labels. Tasks B–D compare each mental-health language category against Normal. This avoids treating Anxiety as a negative class for Depression and avoids merging Suicidal language with Depression.

This task design is more conservative than the earlier coarse high-risk-vs-other formulation. Anxiety can share affective and physiological language with Depression, so using Anxiety as a negative control for Depression can create a clinically ambiguous label contrast. Similarly, Suicidal-language classification should be analyzed separately because crisis terms are both semantically central and potential shortcuts. The one-vs-normal tasks are still benchmark tasks, not diagnosis tasks, but they make the label contrast clearer.

TABLE II
TASK DEFINITIONS AND RATIONALE.

Name	Included labels	Positive class	Primary metric	Rationale
Four-class	Normal, Anxiety, Depression, Suicidal	not applicable	Macro F1	Preserves original dataset labels.
Depression vs Normal	Normal, Depression	Depression	AUROC	Uses Normal as cleaner contrast for Depression.
Suicidal vs Normal	Normal, Suicidal	Suicidal	AUROC	Audits suicidal-language classification separately.
Anxiety vs Normal	Normal, Anxiety	Anxiety	AUROC	Avoids treating Anxiety as a depression negative.

V. METHODS

A. Models and Features

All models used TF-IDF features with unigram and bigram features, lowercase normalization, Unicode accent stripping, `min_df`=3, `max_df`=0.95, and at most 20,000 features for logistic regression, calibrated linear SVM, and multinomial naive Bayes. Stop-word removal was not applied, since function words and short phrases may encode dataset style. Random forest was included as a secondary baseline with at most 10,000 TF-IDF features, `min_df`=5, 60 trees, maximum depth 40, and balanced subsampling.

The primary classical baselines were TF-IDF + Logistic Regression, TF-IDF + Calibrated Linear SVM, and TF-IDF + Multinomial Naive Bayes. The calibrated SVM used Platt-style probability calibration through three-fold internal calibration. Transformer fine-tuning was not run because the local environment was not configured for GPU training and this audit prioritized interpretable baselines.

The model set covers complementary assumptions. Logistic regression provides sparse linear evidence that is easy to inspect through coefficients. The calibrated linear SVM often performs strongly in high-dimensional text spaces while still using a linear decision surface. Multinomial naive Bayes provides a generative bag-of-words baseline common in text classification. Random forest was included only as a secondary baseline because tree ensembles over sparse TF-IDF features can be slower and less interpretable than linear models.

B. Splits, Confidence Intervals, and Calibration

Standard estimates used a stratified 75/25 train/test split with fixed random seed 42. The main classical models also used stratified five-fold cross-validation summaries for full-text conditions. Binary 95% confidence intervals were generated with 300 bootstrap resamples of test predictions. Expected calibration error (ECE) used 10 equal-width probability bins. For binary tasks, metrics include AUROC, AUPRC, accuracy, balanced accuracy, macro F1, F1, precision, sensitivity, specificity, Brier score, ECE, and confusion matrices. For the four-class task, metrics include accuracy, macro F1, weighted F1, per-class precision/recall/F1, macro one-vs-rest AUROC where available, and confusion matrices.

ECE was computed as

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{n} \left| \mathrm{acc}(S_b) - \mathrm{conf}(S_b) \right|, \tag{1}$$

where $B = 10$, S_b is the set of predictions in bin b, $\mathrm{acc}(S_b)$ is the observed positive fraction, and $\mathrm{conf}(S_b)$ is the mean predicted probability. For binary tasks, the Brier score was computed on the positive-class probability.

C. Shortcut Ablations

Direct keyword removal removed exact and morphological label terms: depression, depressed, depressive, anxiety, anxious, suicide, suicidal, suicidality, self harm, self-harm, and mental health. Crisis-term removal removed phrases such as kill myself, end my life, die, death, goodbye, overdose, hurt myself, and no reason to live. Artifact removal removed URLs, handles, forum-style tokens, and repeated platform artifacts. Random deletion removed the same number of tokens as the direct-keyword removal condition when keyword terms were present; this estimates whether performance drops reflect keyword-specific evidence rather than generic information loss.

Length-matched evaluation used five word-count bins and downsampled within each bin to match class counts. Near-duplicate-safe splitting grouped normalized text strings so highly similar examples were less likely to cross train/test boundaries. Length sensitivity was the Pearson correlation between predicted positive probability and word count for binary tasks, and between maximum class probability and word count for the four-class task.

These ablations are intentionally conservative. Removing direct keywords does not prove that remaining performance is clinically meaningful, because models may rely on correlated phrases or stylistic cues. Conversely, a drop after crisis-term removal should not automatically be interpreted as undesirable leakage, because crisis terms may be semantically central to suicidal-language labels. For this reason, the study compares several perturbations rather than treating any single ablation as decisive.

For a condition c and full-text baseline f, shortcut drop was defined as

$$\Delta_c = M_f - M_c, \tag{2}$$

where M is Macro F1 for Task A and AUROC for binary tasks. Robustness retention was

$$R_c = M_c / M_f. \tag{3}$$

Length sensitivity was

$$\rho_\ell = \mathrm{corr}(\hat{p}, \ell), \tag{4}$$

where $\hat{p}$ is the positive-class probability for binary tasks, or maximum class probability for Task A, and ℓ is word count.

D. Shortcut-Risk Rule

Shortcut-risk labels are heuristic and sensitivity-analysis-oriented, not clinically validated. A task/model receives one point for direct-keyword dependence greater than 0.03, one point for crisis-term dependence greater than 0.03, one point for absolute length sensitivity greater than 0.25, one point for near-duplicate-safe performance drop greater than 0.03, and two points when more than 10% of top features are direct leakage, crisis/safety terms, or platform artifacts. Zero points is low risk, one or two points is moderate risk, and three or more points is high risk. If top features contain leakage words or artifacts, low-risk labels should be interpreted cautiously.

VI. Experiments

All tasks were evaluated under full text, direct keyword removal, crisis-term removal, artifact removal, random deletion, early text, late text, length-matched evaluation, and near-duplicate-safe splitting. Large full ablation outputs are stored as CSV files; the main paper reports compact summaries for readability.

The early-text and late-text conditions used the first and last 50% of tokens, respectively. These conditions test whether performance depends disproportionately on lead-in language or later explanatory language. Artifact removal targeted URLs, handles, forum-style strings, and repeated platform tokens. The random deletion control was included because keyword removal changes text length and information content; a keyword-specific effect should exceed the performance drop caused by deleting the same number of arbitrary tokens.

All generated outputs are written to project directories: audit files under `results/audit/`, metrics under `results/metrics/`, predictions under `results/predictions/`, feature analyses under `results/explanations/`, and compact tables under `results/tables/`. This structure makes every manuscript number traceable to a generated artifact rather than hand-entered prose.

A. Reproducibility and Audit Scope

The experimental pipeline is intentionally organized as a reproducible audit rather than a single leaderboard script. Dataset normalization is separated from dataset auditing, model evaluation, table generation, visualization, and macro generation. This separation matters because leakage findings can otherwise be lost when only final metrics are preserved. The audit files record class distributions, missingness, duplicate estimates, word-count summaries, keyword prevalence, artifact prevalence, and top terms by class. The metric files preserve full model-condition results, while the manuscript tables report only compact main-paper summaries.

The scope of the audit is lexical and distributional. It can identify direct label terms, crisis phrases, length differences, approximate near duplicates, URL-like artifacts, and coefficient-level feature categories. It cannot determine whether a post was authored by a person with a clinical condition, whether two samples came from the same user, or whether an expression is literal, quoted, hypothetical, or metaphorical. These boundaries are important: the purpose is to make benchmark evidence more transparent, not to certify the dataset for clinical inference.

VII. Results

A. Dataset Audit Findings

The dataset is moderately imbalanced, with the largest class approximately 3.25 times the smallest class. The audit also found exact duplicates and near-duplicate pairs. Direct keyword, crisis keyword, and artifact prevalence vary by class, indicating that models may benefit from lexical cues that are not necessarily robust across datasets.

The most striking audit pattern is the combination of length and keyword differences. Normal samples are much shorter on median than the three mental-health label classes, while direct label keywords are common in Depression and Anxiety examples. Suicidal examples also show high crisis-term prevalence. These observations do not invalidate the dataset, but they make aggregate performance insufficient as the only evidence of model quality.

This audit motivates the ablation design. If a model assigns higher probabilities to longer posts or to posts containing label names, then held-out performance can remain high even when the model is not learning a general language representation. The duplicate audit similarly motivates duplicate-safe splitting: near-identical examples can make train/test generalization appear stronger than it would be on genuinely independent text.

The audit also clarifies why benchmark labels should be discussed carefully. The Depression, Anxiety, and Suicidal classes are meaningful dataset categories, but their prevalence of direct label and crisis terms means they partly encode how examples were collected or annotated. A classifier can exploit that regularity without understanding mental-health status. The Normal class, by contrast, has shorter median length and lower keyword prevalence, making length and explicit vocabulary potential discriminators. These are precisely the kinds of dataset-level effects that can make public benchmarks easier than real-world screening-support settings.

B. Main Predictive Performance

Table III reports four-class performance. The best four-class model was TF-IDF + Logistic Regression, with macro F1 0.776 and accuracy 0.790. Table IV reports one-vs-normal results. The best AUROC values were 0.984 for Depression vs Normal, 0.988 for Suicidal-language vs Normal, and 0.988 for Anxiety vs Normal. Figure 3 summarizes the main task-level performance.

The binary results are substantially stronger than the four-class result. This is expected because one-vs-normal tasks reduce label ambiguity and ask the model to separate one target label from Normal text. However, this simplification also means binary AUROC should not be compared directly with four-class Macro F1. The purpose of reporting both is to show how the benchmark behaves under label-preserving and cleaner contrastive formulations.

Among the binary tasks, calibrated linear SVM was consistently the strongest full-text model. This pattern is consistent with high-dimensional sparse text classification, where linear margins can separate lexical patterns effectively. The result should still be read as dataset-label predictability. It does not show that the model can infer actual mental-health status, nor that the same performance would hold under a different platform, annotation process, or clinical interview protocol.

The model ranking also illustrates why a simple benchmark table is not enough. A model with excellent AUROC may still have shortcut-sensitive evidence, while a slightly weaker model may be easier to interpret or less dependent on a specific cue. The paper therefore treats performance as one dimension

TABLE III

FOUR-CLASS FULL-TEXT PERFORMANCE. A DASH INDICATES THAT THE METRIC WAS NOT AVAILABLE OR NOT APPLICABLE FOR THE MODEL OUTPUT.

Model	Accuracy	Macro F1	Weighted F1	Macro OVR AUROC	Notes
TF-IDF + Logistic Regression	0.790	0.776	0.788	0.940	One-vs-rest
TF-IDF + Calibrated Linear SVM	0.784	0.768	0.781	0.934	One-vs-rest
TF-IDF + Multinomial Naive Bayes	0.741	0.722	0.740	0.917	One-vs-rest
TF-IDF + Random Forest	0.720	0.695	0.715	0.911	One-vs-rest

TABLE IV

BEST BINARY ONE-VS-NORMAL FULL-TEXT PERFORMANCE. FULL MODEL COMPARISONS ARE RETAINED IN GENERATED CSV OUTPUTS. AUROC INTERVALS ARE BOOTSTRAP 95% CONFIDENCE INTERVALS.

Task	Model	AUROC (95% CI)	AUPRC	F1	Sensitivity	Specificity	Brier	ECE
Suicidal vs Normal	TF-IDF + Calibrated Linear SVM	0.988 [0.986, 0.990]	0.984	0.941	0.928	0.972	0.035	0.011
Anxiety vs Normal	TF-IDF + Calibrated Linear SVM	0.988 [0.986, 0.990]	0.968	0.904	0.880	0.980	0.032	0.016
Depression vs Normal	TF-IDF + Calibrated Linear SVM	0.984 [0.981, 0.986]	0.981	0.933	0.919	0.960	0.045	0.015

of evaluation. Calibration, ablation drops, length sensitivity, duplicate-safe splitting, and feature categories are reported as complementary evidence about reliability.

C. Shortcut Ablation Findings

Table V and Figure 4 summarize shortcut drops for the best model in each task. Direct keyword removal had its largest effect in the four-class task, while binary models retained high AUROC after direct keyword removal. This suggests broader lexical signal remains. However, direct keyword removal alone is not sufficient to establish robustness. High length sensitivity, top-feature leakage/artifact proportion, and duplicate-safe split effects also contribute to the shortcut-risk label. For Suicidal vs Normal, crisis terms are both semantically meaningful and shortcut-like, so interpretation requires particular caution.

D. Calibration Findings

Figure 5 shows reliability curves for the best binary models. Table VI reports Brier score and ECE. Calibration is useful for benchmark assessment, but it is not sufficient evidence for clinical deployment.

The best calibrated binary models had low ECE on the held-out benchmark split. This supports internal benchmark reliability, but calibration can shift under new sources, populations, and annotation rules. The calibration results therefore should be viewed as a property of this dataset split rather than a clinical risk-estimation guarantee.

Calibration also interacts with shortcut sensitivity. A model can be well calibrated on a held-out split drawn from the same benchmark distribution while being poorly calibrated on another dataset where posts are shorter, crisis terms are less explicit, or platform artifacts differ. For this reason, calibration is necessary but not sufficient for trustworthy mental-health text modeling.

E. Interpretable Feature Leakage Findings

The feature analysis is a lexical leakage audit, not a complete clinical explanation of model behavior. Table VII and Figure 6 show category proportions among top TF-IDF logistic-regression features. Examples of leakage or safety terms observed in top-feature files include label words such as depression, depressed, anxiety, suicidal, and suicide; crisis/safety phrases such as die and death; and artifact-like tokens such as http, co, deleted, or removed. Symptom or emotional terms include hopeless, tired, sleep, panic, sad, crying, alone, and lonely. Because many high-weighted TF-IDF features were categorized as unclear lexical signals, the feature analysis should be interpreted as a leakage audit rather than a complete clinical explanation of model behavior.

The large unclear category is itself informative. It indicates that dictionary-based grouping can identify obvious leakage and artifact categories, but cannot fully explain all high-weighted n-grams. Some unclear features may be benign topical terms, some may be dataset-specific phrasing, and some may be indirectly related to symptoms. A complete explanation would require richer linguistic analysis, source metadata, and possibly manual review. For this reason, the feature audit is best interpreted as a conservative screen for obvious problems rather than a comprehensive explanation of model behavior.

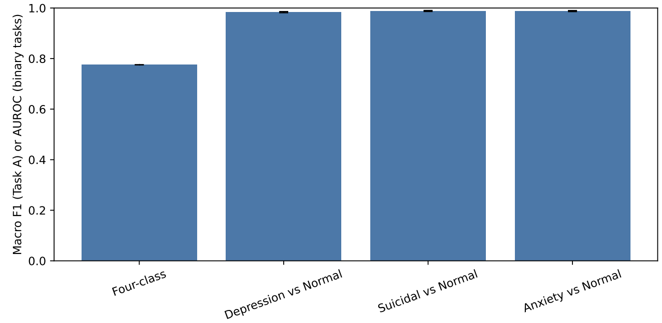

Fig. 3. Main task performance. Bars show the best model for each task: macro F1 for the four-class task and AUROC for the binary one-vs-normal tasks. Error bars show available bootstrap confidence intervals.

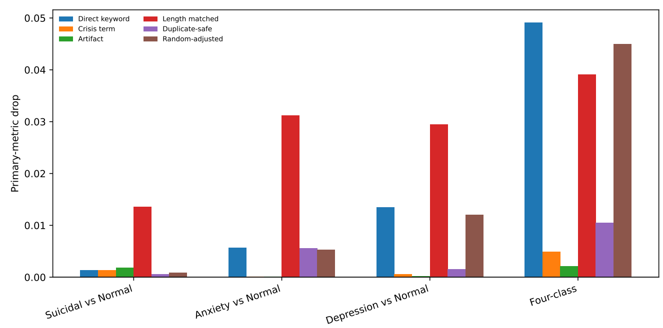

Fig. 4. Shortcut ablation drops. Larger positive values indicate greater sensitivity to the ablation. Random-adjusted drop compares direct keyword removal with matched random token deletion.

VIII. DISCUSSION

Classical TF-IDF models perform strongly on public mental-health text labels. This is encouraging for benchmark modeling but does not imply clinical validity. Keyword removal did not erase performance, suggesting that broader

TABLE V
SHORTCUT ABLATION AND RISK-EVIDENCE SUMMARY FOR THE BEST FULL-TEXT MODEL IN EACH TASK. DROPS ARE RELATIVE TO FULL-TEXT PRIMARY SCORE; RISK POINTS FOLLOW THE HEURISTIC RULE DEFINED IN METHODS.

Task	Best model	Best full-text score	Direct keyword drop	Crisis-term drop	Artifact-removal drop	Length-matched drop	Near-duplicate-safe drop	Random-deletion-adjusted drop	Length sensitivity	Top-feature leakage/artifact %	Risk points	Shortcut risk
Suicidal vs Normal	TF-IDF + Calibrated Linear SVM	0.988	0.001	0.001	0.002	0.014	0.001	0.001	0.425	13.0%	3	high shortcut risk
Anxiety vs Normal	TF-IDF + Calibrated Linear SVM	0.988	0.006	0.000	0.000	0.031	0.006	0.005	0.448	4.0%	1	moderate shortcut risk
Depression vs Normal	TF-IDF + Calibrated Linear SVM	0.984	0.013	0.001	0.000	0.029	0.002	0.012	0.482	10.0%	1	moderate shortcut risk
Four-class	TF-IDF + Logistic Regression	0.776	0.049	0.005	0.002	0.039	0.011	0.045	-0.121	13.5%	3	high shortcut risk

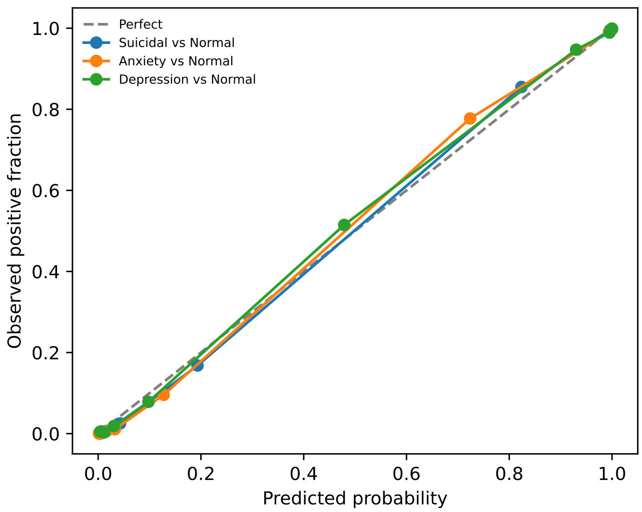

Fig. 5. Calibration curves for the best binary models. Good reliability on this benchmark does not imply clinical deployment readiness.

TABLE VI
FULL-TEXT CALIBRATION METRICS.

Task	Model	Brier	ECE
Depression vs Normal	TF-IDF + Logistic Regression	0.052	0.054
Depression vs Normal	TF-IDF + Calibrated Linear SVM	0.045	0.015
Depression vs Normal	TF-IDF + Multinomial Naive Bayes	0.081	0.081
Depression vs Normal	TF-IDF + Random Forest	0.075	0.068
Suicidal vs Normal	TF-IDF + Logistic Regression	0.044	0.058
Suicidal vs Normal	TF-IDF + Calibrated Linear SVM	0.035	0.011
Suicidal vs Normal	TF-IDF + Multinomial Naive Bayes	0.058	0.074
Suicidal vs Normal	TF-IDF + Random Forest	0.067	0.083
Anxiety vs Normal	TF-IDF + Logistic Regression	0.046	0.073
Anxiety vs Normal	TF-IDF + Calibrated Linear SVM	0.032	0.016
Anxiety vs Normal	TF-IDF + Multinomial Naive Bayes	0.057	0.064
Anxiety vs Normal	TF-IDF + Random Forest	0.063	0.076

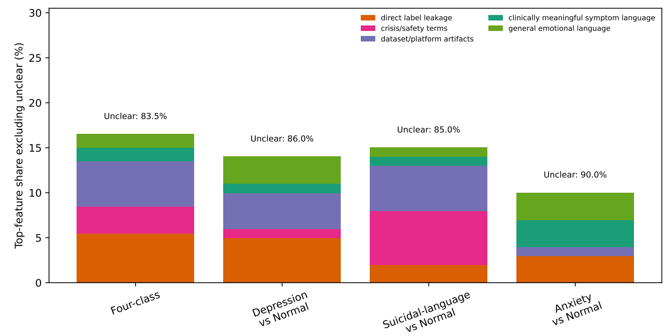

Fig. 6. Top-feature category breakdown, excluding the unclear category from the stacked bars and annotating it separately. Because most top features fall into the unclear category, this analysis is best interpreted as a lexical leakage audit rather than a complete clinical explanation.

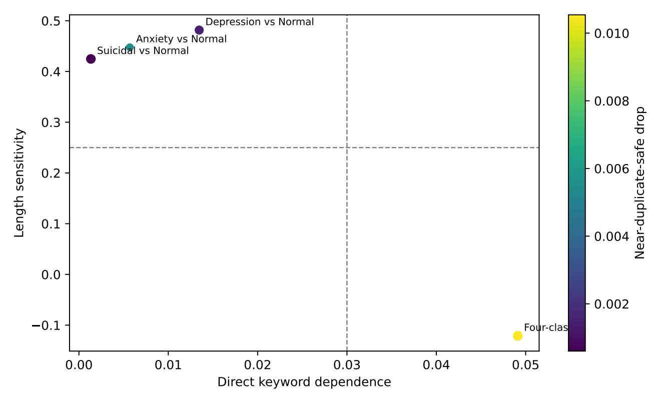

Fig. 7. Shortcut-risk dashboard. The dashed vertical threshold marks direct-keyword dependence of 0.03, and the dashed horizontal threshold marks length sensitivity of 0.25. Point size reflects top-feature leakage/artifact proportion; color reflects near-duplicate-safe drop.

lexical signal remains beyond obvious label words. At the same time, class length differences, keyword prevalence, top-feature leakage, and length sensitivity show that shortcut risks remain substantial.

The results also show why shortcut sensitivity cannot be reduced to a single ablation. A small direct-keyword drop can coexist with high length sensitivity or leakage-heavy top features. Conversely, some direct terms may be semantically meaningful in the label definition, especially in suicidal-language classification. A robust benchmark report should therefore combine perturbation results, duplicate-aware splits, calibration, and feature audits.

The most useful interpretation is balanced. The models are not merely memorizing one or two label words, because performance remains strong under direct keyword removal. At the same time, the models are not shortcut-free, because the audit and feature analysis reveal length effects and leakage-sensitive evidence. This combination is common in health-informatics benchmarks: real signal and artifact signal can coexist. Reporting both is more informative than presenting either an optimistic performance claim or a blanket dismissal of the dataset.

Suicidal-language classification is especially delicate because crisis terms are both semantically meaningful and shortcut-like. Removing such terms may reduce leakage, but those terms are also part of the language being modeled. This makes suicidal-language classification a benchmark audit problem, not a suicide-risk diagnosis system. Public bench-

TABLE VII
FEATURE LEAKAGE SUMMARY. PERCENTAGES ARE AMONG TOP TF-IDF LOGISTIC-REGRESSION FEATURES. A HIGH UNCLEAR-SIGNAL PERCENTAGE WEAKENS INTERPRETABILITY BECAUSE DICTIONARY CATEGORIES DO NOT FULLY EXPLAIN MODEL EVIDENCE.

Task	Condition	Direct label leakage %	Crisis/safety term %	Platform artifact %	Clinically meaningful symptom %	General emotional language %	Unclear lexical signal %
Four-class	Direct Keyword Removed	0.000	3.000	6.000	2.000	2.000	87.000
Four-class	Full Text	5.500	3.000	5.000	1.500	1.500	83.500
Depression vs Normal	Direct Keyword Removed	0.000	1.000	5.000	1.000	3.000	90.000
Depression vs Normal	Full Text	5.000	1.000	4.000	1.000	3.000	86.000
Suicidal vs Normal	Direct Keyword Removed	0.000	6.000	6.000	1.000	1.000	86.000
Suicidal vs Normal	Full Text	2.000	6.000	5.000	1.000	1.000	85.000
Anxiety vs Normal	Direct Keyword Removed	0.000	0.000	1.000	3.000	2.000	94.000
Anxiety vs Normal	Full Text	3.000	0.000	1.000	3.000	3.000	90.000

mark datasets should therefore be audited before being used to claim progress in mental-health AI.

Future work should test cross-dataset generalization, clinical interview datasets, EHR-linked cohorts, and prospectively collected data with stronger consent, sampling, and label provenance. Such work should report calibration, subgroup performance, and leakage analyses alongside aggregate metrics.

A. Implications for Benchmark Reporting

The findings suggest several reporting practices for public mental-health text benchmarks. First, papers should preserve original labels when possible and avoid clinically ambiguous binary mappings unless they are clearly justified. Second, class counts, word-count distributions, duplicate estimates, and keyword prevalence should be reported before modeling. Third, benchmark studies should include at least one perturbation that removes direct label terms and at least one analysis of length sensitivity. Fourth, top features or attribution summaries should be checked for label words, crisis phrases, URLs, usernames, and platform artifacts. Finally, calibration should be reported separately from discrimination because a high AUROC does not guarantee useful probability estimates.

These practices do not make public benchmarks equivalent to clinical datasets. Rather, they make the benchmark's evidentiary limits visible. For BIBM-style health-informatics work, this distinction is important: reproducible public datasets can support method development, but they should not be used to imply clinical readiness without external validation and stronger label provenance.

IX. CLINICAL AND ETHICAL CONSIDERATIONS

This work is not a diagnostic model and is not ready for clinical deployment. The dataset labels come from public online language data and should not be treated as true psychiatric status. Mental-health predictions can cause harm if used without consent, privacy protections, contextual review, and appropriate support pathways. The term Suicidal in this paper refers to a dataset label and is discussed as suicidal-language classification.

False positives can stigmatize users or trigger inappropriate interventions, while false negatives can create false reassurance. Public posts may include quotations, jokes, song lyrics, advocacy language, or third-person discussion. Automated labeling systems can also conflict with user expectations of privacy and consent. For these reasons, this study should be used as an audit of benchmark behavior, not as evidence for operational mental-health screening.

The ethical concern is not only downstream deployment. Even benchmark framing can shape how researchers and readers understand sensitive labels. Treating public text categories as diagnoses can reinforce stigma and obscure the uncertainty of data collection. A conservative audit framing keeps the focus on model behavior under dataset assumptions. It also avoids implying that people represented in public text can or should be classified without consent or contextual review.

X. LIMITATIONS

The dataset lacks clinician-confirmed diagnoses and detailed sampling provenance. The annotation process may not reflect clinical assessment. Near-duplicate detection is approximate and may miss paraphrases. Length matching reduces but does not eliminate confounding. Dictionary-based feature categories are incomplete, leaving many high-weighted features as unclear lexical signals. Transformer models and cross-dataset validation were not run in this local study.

The dataset also lacks user identifiers, so user-level splitting could not be performed. If multiple samples come from the same author, standard train/test splits may overestimate generalization. Demographic subgroup analyses were unavailable. The keyword dictionaries used for ablation and feature categorization are necessarily incomplete and English-centric. Future work should repeat this audit across datasets and include clinical interview or EHR-linked cohorts when appropriate governance and consent are available.

XI. CONCLUSION

This study presents a conservative leakage and shortcut-sensitivity audit of public mental-health text classification. Strong TF-IDF performance is reproducible, but lexical leakage, length sensitivity, and artifact-sensitive features limit interpretation. Public mental-health benchmarks should be accompanied by dataset audits, shortcut ablations, calibration checks, and lexical feature review before they are used to support claims about trustworthy mental-health AI.

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